

# UDIT MISHRA

**Generative AI Engineer | LLM Agents, RAG, Machine Learning, Voice AI**

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## SUMMARY

Generative AI Engineer with 3 years of hands-on experience designing and deploying production machine learning and LLM systems, including multi-agent pipelines (LangChain, LangGraph), retrieval-augmented generation (RAG) with vector databases (Qdrant, FAISS), embeddings-based classification, and voice-enabled conversational AI (STT/TTS). Comfortable owning the full ML/AI lifecycle: data pipelines, model integration and evaluation, feature engineering, FastAPI backends, PostgreSQL, Redis, Neo4j, Docker deployment, and multi-provider LLM orchestration across OpenAI, Anthropic, Gemini, and open-source models.

## TECHNICAL SKILLS

**Machine Learning:** Model Evaluation, Feature Engineering, Embeddings, Vector Similarity Search, Classification, Scikit-learn, NumPy, Pandas, Model Fine-Tuning Fundamentals

**LLM and Agentic AI:** LangChain, LangGraph, LlamaIndex, Multi-Agent Orchestration, Prompt Engineering, RAG Architecture, Tool/Function Calling

**Voice and NLP:** STT/TTS Integration, Real-time Voice Streaming, Embeddings, HuggingFace, NLP

**Python:** Python 3.10+, OOPs, FastAPI, Async Programming, Jupyter Notebooks

**Vector and Databases:** Qdrant, FAISS, PostgreSQL, Redis, MySQL, Neo4j

**DevOps and APIs:** Docker, Docker Compose, CI/CD (GitHub), JWT Auth, Alembic, REST APIs, Node.js/Express

**Frontend:** React.js, Next.js, TypeScript, Angular 19

## EXPERIENCE

### WebbyButter Technologies Pvt. Ltd., Vadodara

#### Artificial Intelligence Engineer

*Feb 2024 – Present*

- Architected and deployed multi-agent LLM/ML pipelines using LangGraph and LangChain, integrating 7+ model providers (OpenAI, Anthropic, Gemini, Mistral, Cohere, Ollama, HuggingFace) for production workloads.
- Delivered enterprise AI/ML solutions for production clients including Evenuefy and PharmaSuper.ai, building end-to-end machine learning applications, RAG pipelines, backend APIs, cloud deployments, and production support.
- Built RAG architectures and embedding-based retrieval pipelines in Python and Jupyter, applying feature engineering and similarity scoring to orchestrate retrieval and generation across multiple LLM providers.
- Developed and deployed containerized FastAPI applications (Docker Compose) covering REST APIs, embedding services, and vector storage layers.
- Published open-source Python SDK `Ai_Gateway` on PyPI, providing fallback routing, latency tracking, and response quality scoring for multi-provider LLM deployments.

#### Software Engineer

*Aug 2023 – Feb 2024*

- Built and maintained JWT-secured REST APIs with Alembic migrations on PostgreSQL for production backend services.
- Maintained CI/CD workflows on GitHub and deployed to Docker/VM environments across testing, staging, and production.

#### Frontend Developer Intern | Top-Notch IT Consultant Pvt. Ltd., Vadodara

*May 2023 – Jun 2023*

- Built responsive SPAs using React.js with reusable component architecture, API integration, and performance optimization via Hooks and lazy loading.

## KEY PROJECTS

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### PharmaSuper.ai – AI/ML Assistant for the Pharmaceutical Industry

- Built enterprise AI/ML features for a pharmaceutical platform using machine learning models and Retrieval-Augmented Generation.
- Developed backend APIs, vector search pipelines, and semantic document retrieval for pharmaceutical knowledge.
- Integrated multiple AI models, optimized prompt engineering, and improved model response accuracy for production use cases.

### Evenuefy – AI/ML-Powered Event Management Platform

- Developed AI/ML-powered features for a production event management platform supporting virtual, hybrid, and in-person events.
- Built and integrated LLM-powered conversational assistants for event information, attendee support, and platform navigation.
- Developed scalable backend APIs and RAG-based retrieval workflows for intelligent document and knowledge search.
- Managed production deployments, bug fixes, CI/CD pipelines, and cloud infrastructure to ensure reliable releases.

### Conversational AI Voice Assistant – Full-Stack

- Built a real-time voice and text AI interface with streaming responses, transcript export, and browser audio API integration (STT/TTS pipeline) using Next.js, TypeScript, and Browser Audio APIs.

### LoopMind Memory – Production AI Memory Backend

- Built a hierarchical memory system with ANN/HNSW retrieval achieving sub-100ms lookup, containerized via Docker Compose with PostgreSQL, Redis, Qdrant, Neo4j, and FastAPI.
- Implemented tamper-evident append-only storage with signed audit ledger, JWT auth, and a complete test suite.

### SQL Conversational Agent – NL2SQL and RAG Pipeline

- Engineered a natural language to SQL agent with query validation, sandboxed execution, and a feedback/rating system using OpenAI GPT-4o and Qdrant-powered schema retrieval.

## OPEN SOURCE

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- `Ai_Gateway` ([pypi.org/project/ai-gateway/](https://pypi.org/project/ai-gateway/)): Published Python SDK for LLM observability, fallback routing, and response quality evaluation across OpenAI, Anthropic, Gemini, and OSS models.

## EDUCATION

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**Master of Computer Applications (MCA), Information Technology** – SVIT, Vasad (Expected Apr 2027)      2025 – 2027

**Bachelor of Computer Applications (BCA), Information Technology** – Parul University, Vadodara — CGPA 7.66      2022 – 2025

## CERTIFICATIONS

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- Artificial Intelligence, Great Learning (Neural Networks, NLP, ML Fundamentals)
- Python Programming, IIT Bombay